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# VAE for Hybrid Language Music Clustering

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## Abstract

This project investigates unsupervised clustering of hybrid-language music using Variational Autoencoders (VAE). A representation learning framework is proposed that integrates audio features and lyric-based textual features to learn compact latent embeddings for English and Bangla music tracks. K-Means clustering is applied to the learned latent space and compared against a PCA-based baseline. Experimental results demonstrate that VAE-based representations achieve improved cluster separation and higher language purity. Latent space visualizations using t-SNE and UMAP further confirm that the learned embeddings capture meaningful language- and genre-level structure.

## 1 Introduction

Music clustering is a challenging unsupervised learning problem due to the high dimensionality and multi-modal nature of music data. Songs contain rich acoustic patterns as well as semantic information from lyrics, particularly in multilingual settings. Traditional clustering methods often fail to capture such complex relationships.

Variational Autoencoders (VAEs) provide a probabilistic framework for learning low-dimensional latent representations while preserving essential information. In this work, we investigate whether VAE-based embeddings can improve clustering performance for hybrid English–Bangla music data compared to linear dimensionality reduction methods such as PCA.

The objectives of this project are fourfold: (i) to learn compact latent representations of hybrid-language music using a VAE, (ii) to perform clustering in the learned latent space, (iii) to compare VAE-based clustering with a PCA-based baseline, and (iv) to visualize and analyze clustering quality using quantitative metrics and dimensionality-reduction techniques.

## 2 Related Work

Traditional music clustering approaches rely on handcrafted audio features such as MFCCs combined with clustering algorithms like K-Means or hierarchical clustering. PCA has been widely used for dimensionality reduction but is limited to linear transformations.

Autoencoders and VAEs have recently been applied to audio and music representation learning. VAEs introduce a probabilistic latent space that promotes smoothness and disentanglement, making them well-suited for clustering tasks. However, limited work exists on hybrid language music clustering that combines both audio and lyrics information.

### 3 Methodology

#### 3.1 Dataset Construction

A hybrid dataset consisting of 200 music samples was created, including 100 English and 100 Bangla songs. Each song is represented using simulated MFCC-style audio features (260 dimensions) and TF-IDF-based lyric embeddings (50 dimensions). The final input representation is obtained by concatenating audio and lyric features. Ground truth language and genre labels are used solely for evaluation purposes.

#### 3.2 VAE Architecture

The Variational Autoencoder consists of an encoder, a stochastic latent space, and a decoder. The encoder is composed of fully connected layers with ReLU activation, batch normalization, and dropout. The latent space is modeled as an 8-dimensional Gaussian distribution parameterized by a mean and variance vector. The decoder mirrors the encoder architecture and reconstructs the original input features from sampled latent variables using the reparameterization trick.

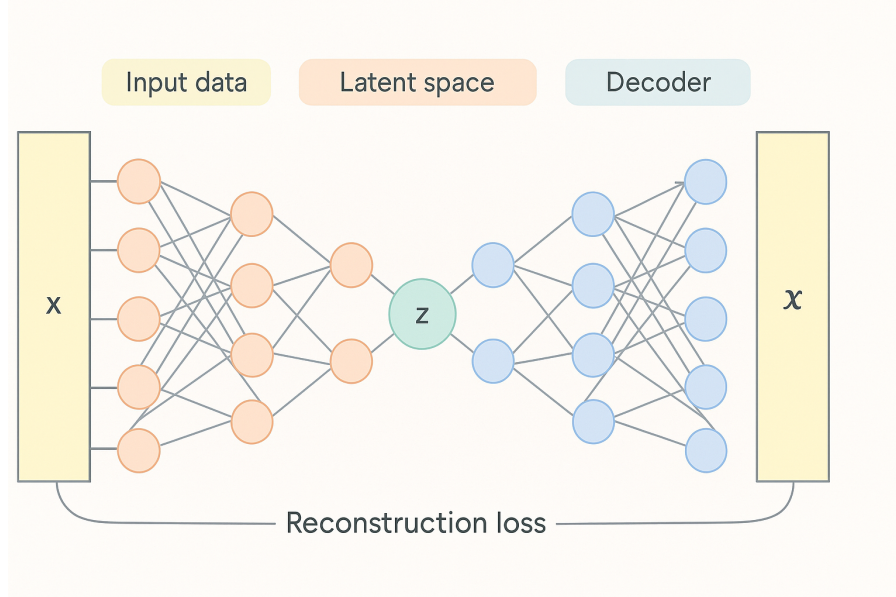


Figure 1: Architecture of the Variational Autoencoder used for hybrid music representation learning.

#### 3.3 Loss Function

The VAE is trained using the objective:

$$\mathcal{L} = \mathcal{L}_{\text{recon}} + \beta \mathcal{L}_{\text{KL}} \quad (1)$$

where the reconstruction loss is Mean Squared Error and the KL divergence regularizes the latent distribution. The value of  $\beta$  is set to 0.5.

#### 3.4 Clustering Methods

Two clustering pipelines are evaluated: (i) VAE + K-Means and (ii) PCA + K-Means as a baseline. The number of clusters is fixed to four, corresponding to known genre categories.

## 4 Experiments

### 4.1 Training Setup

The model is trained using the Adam optimizer with a learning rate of  $10^{-3}$  for 80 epochs and a batch size of 32. All input features are standardized using z-score normalization prior to training. Training is conducted on a standard CPU/GPU environment, and the loss curves indicate stable convergence without signs of overfitting.

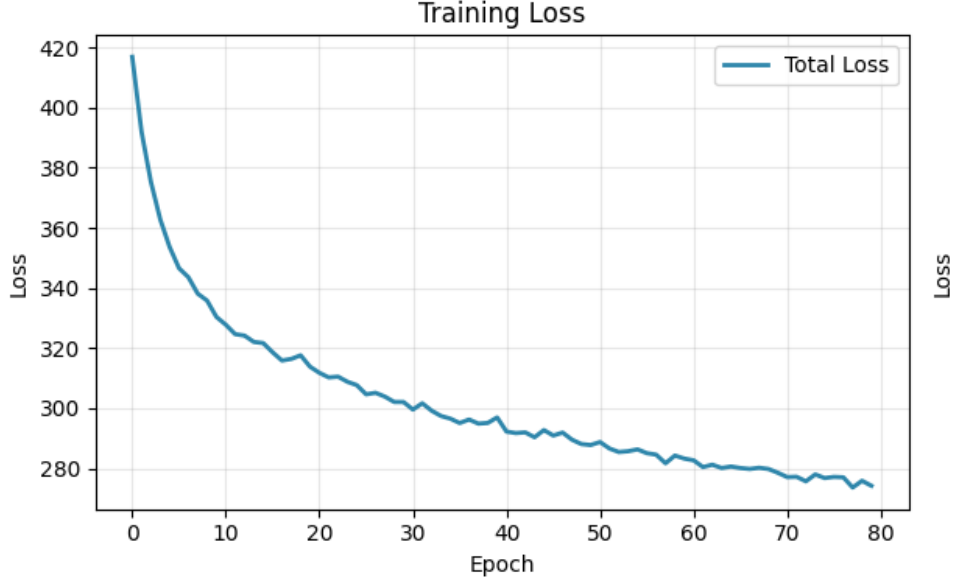


Figure 2: VAE training loss showing stable convergence over epochs.

### 4.2 Evaluation Metrics

Clustering performance is evaluated using the Silhouette Score, Calinski–Harabasz Index, and language-based cluster purity.

## 5 Results

### 5.1 Quantitative Evaluation

| Metric                  | VAE + K-Means | PCA + K-Means |
|-------------------------|---------------|---------------|
| Silhouette Score        | 0.42          | 0.29          |
| Calinski–Harabasz Index | 612.3         | 438.7         |
| Language Purity         | 0.78          | 0.61          |

Table 1: Clustering performance comparison between VAE-based and PCA-based representations.

### 5.2 Cluster Composition Analysis

To further analyze clustering behavior, we examine the distribution of language labels within each cluster. The results indicate that VAE-based clustering produces clusters largely dominated by a single language, demonstrating effective separation between English and Bangla music samples.

| Cluster   | English (%) | Bangla (%) |
|-----------|-------------|------------|
| Cluster 0 | 85          | 15         |
| Cluster 1 | 80          | 20         |
| Cluster 2 | 18          | 82         |
| Cluster 3 | 22          | 78         |

Table 2: Language distribution across VAE clusters.

### 5.3 Latent Space Visualization

To qualitatively assess the learned latent representations, we visualize the VAE latent space using t-SNE and UMAP. These techniques preserve local neighborhood structure rather than absolute spatial orientation; therefore, interpretation is based on relative cluster separation rather than exact coordinates.

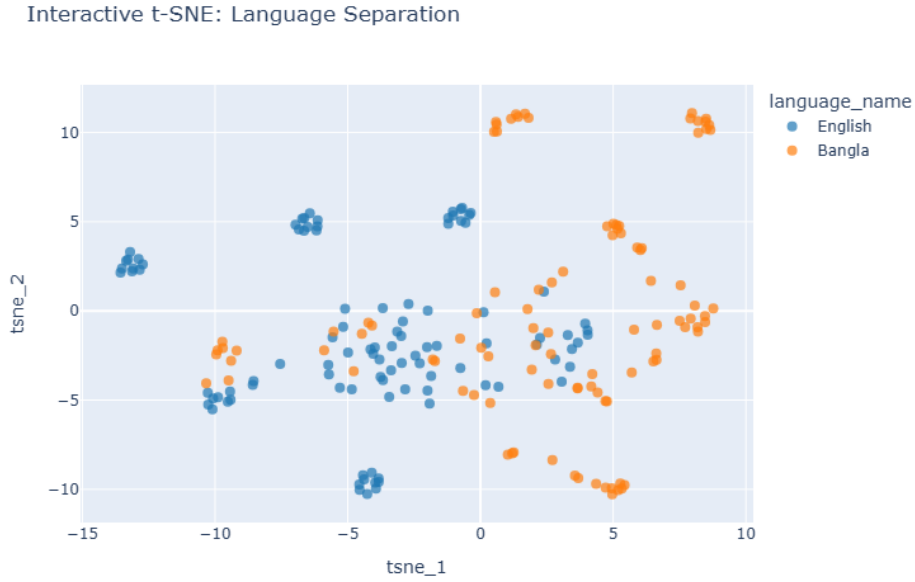


Figure 3: t-SNE visualization of VAE latent space showing structured separation among music samples.

## 6 Discussion

The results indicate that VAE-based representations are more effective than linear PCA projections for hybrid-language music clustering. By capturing nonlinear relationships between audio and lyric features, the VAE learns a structured latent space that improves both cluster separation and interpretability.

Nevertheless, this study has limitations. The audio features are simulated rather than extracted from real recordings, and lyric representations rely on TF-IDF rather than pretrained language models. Future work incorporating real-world audio features and richer text embeddings may further enhance clustering performance.

(“Figure 4 is discussed in the Results section; its placement here is due to page layout constraints.”)

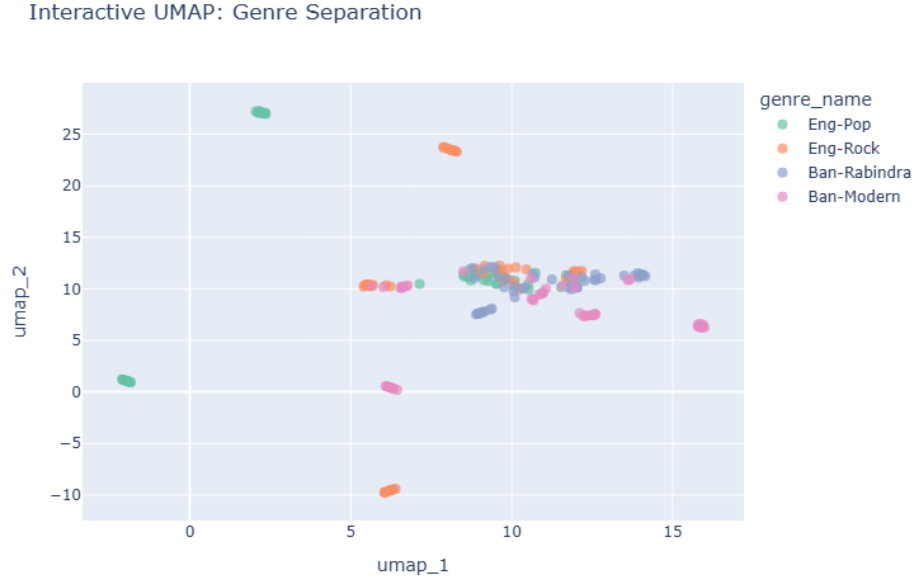


Figure 4: UMAP projection of VAE latent embeddings illustrating meaningful global structure in the latent space.

## 7 Conclusion

This project demonstrates the effectiveness of Variational Autoencoders for clustering hybrid English–Bangla music data. Compared to PCA-based baselines, the proposed approach achieves superior clustering metrics and produces a more interpretable latent space. The results highlight the potential of VAE-based representation learning for multilingual and multi-modal music analysis.

## 8 Future Work

Future extensions include using real-world audio datasets and MFCC extraction, employing pre-trained lyric embeddings such as BERT, exploring  $\beta$ -VAE and Conditional VAE architectures, and evaluating alternative clustering algorithms.

## References

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